

Subqueries

Finding the average amount paid by the top 5 customers.

```
SELECT AVG(total_amount_paid) AS avg_amount_paid
FROM
(SELECT A.customer_id,
B.first_name,
B.last_name,
D.city,
E.country,
SUM(amount) AS total_amount_paid
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
'Pingxiang','Sivas','Celaya','So Leopoldo')
AND country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
ORDER BY SUM(amount) DESC
LIMIT 5) AS average
```

The screenshot shows a SQL query editor with the following query in the 'Query' tab:

```
1 SELECT AVG(total_amount_paid) AS avg_amount_paid
2 FROM
3 (SELECT A.customer_id,
4 B.first_name,
5 B.last_name,
6 D.city,
7 E.country,
8 SUM(amount) AS total_amount_paid
9 FROM payment A
10 INNER JOIN customer B ON A.customer_id = B.customer_id
11 INNER JOIN address C ON B.address_id = C.address_id
12 INNER JOIN city D ON C.city_id = D.city_id
13 INNER JOIN country E ON D.country_id = E.country_id
14 WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
15 'Pingxiang','Sivas','Celaya','So Leopoldo')
16 AND country IN ('India','China','United States','Japan','Mexico','Brazil',
17 'Russian Federation','Philippines','Turkey','Indonesia')
18 GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
19 ORDER BY SUM(amount) DESC
20 LIMIT 5) AS average
```

The 'Data Output' tab shows the result of the query:

avg_amount_paid
107.35400000000000

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:03.888'.

Finding out how many of the top 5 customers I identified in step 1 are based within each country.

```

SELECT D.country,COUNT(DISTINCT A.customer_id) AS all_customer_count,
COUNT(DISTINCT top_5_customers.customer_id) AS top_5_customers
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
LEFT JOIN
(SELECT A.customer_id,
B.first_name,
B.last_name,
D.city,
E.country,
SUM(amount) AS total_amount_paid
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
'Pingxiang','Sivas','Celaya','So Leopoldo')
AND country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
ORDER BY SUM(amount) DESC
LIMIT 5) AS top_5_customers ON top_5_customers.country = D.country
GROUP BY D.country
ORDER BY top_5_customers DESC,all_customer_count DESC

```

Query	Query History
1	SELECT D.country,COUNT(DISTINCT A.customer_id) AS all_customer_count,
2	COUNT(DISTINCT top_5_customers.customer_id) AS top_5_customers
3	FROM customer A
4	INNER JOIN address B ON A.address_id = B.address_id
5	INNER JOIN city C ON B.city_id = C.city_id
6	INNER JOIN country D ON C.country_id = D.country_id
7	LEFT JOIN
8	(SELECT A.customer_id,
9	B.first_name,
10	B.last_name,
11	D.city,
12	E.country,
13	SUM(amount) AS total_amount_paid
14	FROM payment A
15	INNER JOIN customer B ON A.customer_id = B.customer_id
16	INNER JOIN address C ON B.address_id = C.address_id
17	INNER JOIN city D ON C.city_id = D.city_id
18	INNER JOIN country E ON D.country_id = E.country_id
19	WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
20	'Pingxiang','Sivas','Celaya','So Leopoldo')
21	AND country IN ('India','China','United States','Japan','Mexico','Brazil',
22	'Russian Federation','Philippines','Turkey','Indonesia')
23	GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
24	ORDER BY SUM(amount) DESC
25	LIMIT 5) AS top_5_customers ON top_5_customers.country = D.country
26	GROUP BY D.country
27	ORDER BY top_5_customers DESC,all_customer_count DESC

country	all_customer_count	top_5_customers
Mexico	30	2
India	60	1
United States	36	1
Turkey	15	1

Total rows: 108 Query complete 00:00:01.702